BMO
Financial Group
Finance Research
and Trading Lab

Rotman

# Liquidity Risk Case

## Rotman International Trading Competition 2026

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## OVERVIEW

The Liquidity Risk Case challenges participants to put their critical thinking and analytical abilities to test in an environment that requires them to evaluate the liquidity risk associated with tender offers. Participants will be faced with multiple tender offers throughout the case. This will require participants to make rapid judgments on the profitability, subsequent acceptance and execution, or rejection, of each offer. Profits can be generated by taking advantage of price differentials between market prices and prices offered in the private tenders. Once any tender has been accepted, participants should aim to efficiently close out their large positions to maximize returns and minimize liquidity and market risks.

## DESCRIPTION

There will be 2 heats and teams will allocate 2 team members for each heat. Each participant may compete in only one of the 2 heats. Each heat will consist of 5 independent sub-heats, with each sub-heat representing one month of calendar time. Each heat may involve up to four stocks with different volatility and liquidity characteristics.

| Parameter | Value |
|---|---|
| Number of sub-heats | 5 |
| Trading time per sub-heat | 10 minutes (600 seconds) |
| Calendar time per sub-heat | 1 month (20 trading days) |

Tender offers will be generated by computerized traders and distributed at random intervals to random participants. Participants must subsequently evaluate the profitability of these tenders when accepting or bidding on them. Order submission using the RIT API will be enabled. Data retrieval via Real-time Data (RTD) Links and the RIT API will be enabled.

## MARKET DYNAMICS

There are five sub-heats, each with unique market dynamics and parameters as shown below.

| SUB-HEAT 1 | | |
|---|---|---|
| **Securities** | **RITC** | **COMP** |
| **Start Price** | $50 | $40 |
| **Commissions** | $0.02 | $0.02 |
| **Tender Offer Window** | 30 Seconds | 30 Seconds |
| **Volatility** | Low | Medium |
| **Liquidity** | Medium | High |

| SUB-HEAT 2 | | |
|---|---|---|
| **Securities** | **TRNT** | **MTRL** |
| **Start Price** | $15 | $30 |
| **Commissions** | $0.01 | $0.01 |
| **Tender Offer Window** | 30 Seconds | 30 Seconds |
| **Volatility** | High | Low |
| **Liquidity** | Medium | Low |

| SUB-HEAT 3 | | | |
|---|---|---|---|
| **Securities** | **BLU** | **RED** | **GRN** |
| **Start Price** | $10 | $25 | $30 |
| **Commissions** | $0.04 | $0.03 | $0.02 |
| **Tender Offer Window** | 30 Seconds | 30 Seconds | 30 Seconds |
| **Volatility** | High | Low | Medium |
| **Liquidity** | High | Medium | Medium |

| SUB-HEAT 4 | | | |
|---|---|---|---|
| **Securities** | **WDY** | **BZZ** | **BNN** |
| **Start Price** | $12 | $18 | $24 |
| **Commissions** | $0.02 | $0.02 | $0.03 |
| **Tender Offer Window** | 20 Seconds | 20 Seconds | 20 Seconds |
| **Volatility** | Medium | High | Medium |
| **Liquidity** | High | Medium | Medium |

| SUB-HEAT 5 | | | | |
|---|---|---|---|---|
| **Securities** | **VNS** | **MRS** | **JPTR** | **STRN** |
| **Start Price** | $20 | $75 | $35 | $50 |
| **Commissions** | $0.02 | $0.02 | $0.02 | $0.02 |
| **Tender Offer Window** | 20 Seconds | 20 Seconds | 20 Seconds | 20 Seconds |
| **Volatility** | High | Medium | Low | High |
| **Liquidity** | Medium | High | Medium | Medium |

During each sub-heat, participants will occasionally receive one of three different types of tender offers: private tenders, competitive auctions, and winner-take-all tenders. Tender offers are generated by the server and randomly distributed to random participants at different times. Over the course of each sub-heat, every participant will receive the same number of tender offers and face comparable profit potential, although offer prices and quantities may vary. As a result, no participant has a systematic advantage in overall profit opportunities. No trading commission is charged on tender offers.

Private Tenders are routed to individual participants and are offers to purchase or sell a fixed volume of stocks at a fixed price. The tender price is influenced by the current market price.

Competitive Auction offers are sent to all participants at the same time. Participants will be required to determine a competitive, yet profitable, price to submit for a given volume of stocks from the auction. Any participant that submits an order that is better than the base-line reserve price (hidden from participants) will automatically have their order filled, regardless of other participants' bids or offers. If accepted, the transactions will occur at the price that the participant submitted.

Winner-Take-All Tenders request participants to submit bids or offers to buy or sell a fixed volume of stocks. After all prices have been received, the tender is awarded to the participant with the single highest bid or single lowest offer. The winning price, however, must meet a base-line reserve price (hidden from participants). If no bid or offer meets the reserve price, then the trade will not be awarded to anyone (e.g. if all participants bid $2.00 for a $10.00 reserve price stock, nobody will win).

## CALCULATION OF THE PROFIT OR LOSS OF TRADERS

The prices generated by RIT for this case follow a random-walk process using a return drawn from a normal distribution with a mean of zero. That is, at any point in the case simulation, the probability that the price will go up is equal to the probability that the price will go down. This means that participants cannot predict the future price of the stocks without "taking a bet". Therefore, the RITC scoring committee will consider trading stocks for reasons other than reducing the exposure associated with accepting a tender offer to be equivalent to speculating (taking a bet) on the price movement. These types of trades will be flagged as "speculative trades".

Participants will have time to think about the tender offer before they choose to accept it or decline it and the time may be different for each security. For example, one may receive a tender offer at time $t = 0$ and will have until $t = 30$ to decide whether to accept or decline. Any trades for that security made by a participant during this time without accepting or declining the tender offer will be considered as "*front-running*"[1] since the participant had the advance knowledge of a pending institutional order.

This case is designed to reward teams for identifying tender opportunities, accepting them, and efficiently closing out resulting positions, while managing liquidity, market, and execution risk. To reinforce this objective, the RIT system applies automatic, real-time penalties to trading activity flagged as speculative or front-running, and to any tender-related exposure that is not properly closed out by the end of the sub-heat.

[1] Front-running is the unethical and illegal practice of trading a security for your own account while taking advantage of the information contained in the pending orders from your institutional clients.

**Automatic fines (real-time in RIT)**

The system applies the following fine schedule, aggregated across all stocks for the entire iteration:

- $1 per share for flagged speculative/front-running shares up to 5,000 shares
- $2 per share for any additional flagged shares beyond 5,000 shares
- $10 per share for any shares that remain uncovered at the end of the iteration

These fines are incorporated directly into your real-time P&L in RIT, so your displayed performance reflects both trading outcomes and any applicable penalties throughout the case.

## TRADING LIMITS AND TRANSACTION COSTS

Each participant will be subject to gross and net trading limits of 250,000 and 150,000 shares, respectively. The gross trading limit reflects the sum of the absolute values of the long and short positions across all stocks, while the net trading limit reflects the sum of long and short positions such that short positions negate any long positions. Trading limits will be strictly enforced and participants will not be able to exceed them.

The maximum order size is 10,000 shares and commissions are specified in the table above.

## POSITION CLOSE-OUT

Any open position will be closed out at the end of each sub-heat based on the last traded price. This includes any long or short position open in any security. Computerized market makers will increase the liquidity in the market towards the end of trading to ensure the closing price cannot be manipulated.

## KEY OBJECTIVES

**Objective 1**

Evaluate the profitability of tender offers by analyzing market liquidity. Participants should accept tenders that are expected to generate positive profits while rejecting unattractive tender offers.

**Objective 2**

Submit competitive, yet profitable, bids and offers for competitive auction and winner-take-all tenders to maximize potential profits while managing liquidity and market risk.

**Objective 3**

Use a combination of limit orders, market orders, and marketable limit orders to mitigate liquidity and market risks from holding open positions. There is a chance that the market may move away from your transaction prices, so maintaining large open positions may result in losses.